Coffee Defects Detection Based on Green Bean Images Using YOLO Architecture

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***Abstract*— The quality of coffee beans significantly impacts the overall value of coffee production, and detecting defective beans is a crucial step in ensuring high standards. The traditional coffee bean quality assessment process is time-consuming and subjective, relying heavily on the expertise of a Q grader. However, there is an alternative approach using deep learning technology to develop more portable and trainable tools that can achieve high accuracy. This study compares the performance of two deep learning-based object detection methods, YOLOv8 and YOLOv10, for detecting defects in green Arabica coffee beans, categorized into two different types of primary defects. The research methodology involved collecting data, preprocessing, designing models, training, and testing. The dataset consists of 700 images of coffee beans divided into two classes, brown beans and dried cherries. YOLOv8n achieved a test accuracy of mAP50 0.995 and mAP50-95 0.875, while YOLOv10n attained a test accuracy of mAP50 0.992 and mAP50-95 0.859. In terms of computational efficiency, YOLOv8n completed training in 0.499 hours, whereas YOLOv10n finished in 0.479 hours. Despite the marginal difference in accuracy, both models demonstrate high potential for real-time detection in the coffee industry, with YOLOv8n slightly outperforming YOLOv10n in accuracy, but YOLOv10n being slightly faster. These results suggest that both methods can be effectively utilized for defect detection, offering scalable solutions for the coffee processing industry, particularly in the sorting process, to ensure the production of high-quality, valuable coffee beans.**

Keywords*— *coffee defects, YOLOv8, YOLOv10, object detection, deep learning

I. INTRODUCTION (*HEADING 1*)

The coffee industry contributes to economic restructuring in many countries [1], [2]. As a result, coffee is considered one of the most globally traded commodities and has become a major export of economic importance for developing countries such as Brazil, Vietnam, Colombia, Indonesia, and Ethiopia [3]. Various types of local coffee, such as arabica, are widely recognized globally, with demand increasing each year [4]. This demand is largely driven by the quality of coffee beans, which is a key factor in consumer selection [5]. Consequently, there is a growing need for an accurate, scalable, and cost-effective solution to automate defect detection in coffee beans [6].

The Specialty Coffee Association (SCA) provides one of the most widely recognized methods for classifying green coffee beans [7]. Traditionally, quality assessment and classification of coffee beans rely on visual inspection, which requires significant time and effort and is prone to human error. This dependence on manual inspection results in high labor costs, and consistent quality control is challenging due to various objective factors, such as the difficulty of identifying small defects with the human eye and the impact of inspectors' physical conditions on sample evaluation. Consequently, there is a need for a more objective, automated, and efficient method to accurately assess and classify the quality of coffee beans [1].

In recent years, advances in deep learning have provided alternative approaches to automating quality control processes, including defect detection in agricultural products, such as coffee bean classification [8]. One widely used solution is computer vision [9]. In computational applications, two leading supervised techniques often used are machine learning and deep learning [10]. A popular method within deep learning is the Convolutional Neural Network (CNN) [11]. This deep learning approach, when applied to the coffee bean sorting problem, can achieve an accuracy of 94.63% through automatic feature extraction [12]. However, multi-object identification requires more sophisticated models or algorithms, such as YOLO.

You Only Look Once (YOLO) is a popular CNN-based model for object detection and is often used today [13]. The implementation of 3 versions of YOLO to inspect coffee defects achieved the highest mAP value of 0.993% [14]. Another implementation used YOLOv8 for real time classification of coffee beans. The experimental results improved accuracy and efficiency in the task of classifying coffee bean quality on nine different types of coffee beans [1]. The development of YOLO has reached various versions, until now the YOLO version which is claimed to achieve the most advanced performance and efficiency for end-to-end object detection in real time is YOLOv10 [15].

With the ongoing advancements in object detection technology, particularly in deep learning models like YOLO, there is potential to further enhance the accuracy and efficiency of defect detection in coffee beans. The demand for high-quality coffee has increased globally, driving the need for efficient, accurate, and scalable methods to assess coffee bean quality. This study aims to develop a reliable automated defect detection system for green Arabica coffee beans by leveraging the latest YOLO-based object detection architectures, YOLOv8 and YOLOv10. Specifically, this research focuses on comparing these two deep learning models in terms of accuracy and computational efficiency under challenging conditions, with the goal of identifying which model offers the most

practical solution for real-time defect detection in coffee processing applications.

The key contributions of this research are as follows:

1. Comparative Analysis: Evaluation of YOLOv8 and YOLOv10 models in detecting two types of primary coffee bean defects under challenging conditions.
2. Performance Metrics: Detailed comparison of model performance using standard metrics such as Precision, Recall, mAP (Mean Average Precision), and inference speed.
3. Scalability and Practicality: Investigation of the suitability of both models for deployment in real-time coffee sorting applications, focusing on trade-offs between accuracy and computational resources.

This study aims to provide insights into which YOLO model is more effective for real-world coffee defect detection, potentially helping the coffee industry adopt more efficient and cost-effective quality control systems.

II. Literature Review

A. Deep Learning Based Assessment Technology

Research conducted by Huang et al., to solve the problem of manual coffee defect inspection by classifying green coffee beans in realtime using the Convolutional Neural Network (CNN) method. This research uses coffee bean data categorized into two classes, namely good beans and bad beans, which are augmented with the same proposition. The implementation of deep learning and CNN in this research is able to extract features according to the characteristics of the coffee bean image automatically. The accuracy results reached 94.63% accuracy. However, the algorithm used in this study is limited to the scope of single image classification [12].

Another research developed a coffee supply chain tracking system using blockchain technology and machine learning algorithms to ensure the quality of coffee bean production. This research selects the best model from the YOLOv5m, YOLOv6m and YOLOv7 algorithms. YOLOv5m outperforms other models with a mAP of 99.4% and is able to identify poor quality coffee beans. Proving that YOLO performance can be applied to object detection [14].

Classifying the quality of coffee beans in real-time with the integration of cloud-based systems. Image processing and data augmentation techniques are used to handle coffee bean image data. the model is trained using YOLOv8 to classify coffee beans. then an application is developed for real-time coffee bean recognition through video and image streaming using React Native, NodeJS, and Python. Experimental results provided empirical evidence that the system improved accuracy and efficiency in the coffee bean quality classification task [1].

The improved YOLOv8 implementation successfully detects bluberry aiming to improve bluberry detection accuracy under picking conditions, overcoming challenges such as customary fruit spacing, color similarity between unripe fruit and leaves, and occlusion of fruit by leaves. The experimental results achieved a precision value of 84.6%; recall 91.3%; and mAP value of 94.1% [16].

B. Arabica Coffee

Coffee is grown in 52 mostly low-income countries, and 70% of this coffee production figure comes from the Coffee arabicae species [17]. Arabica coffee is one type of plantation crop that grows and develops in Indonesia [18]. Arabica coffee has a distinctive aroma and distinctive taste and has good content for the body such as antioxidants and other bioactive compounds. This coffee plant will grow optimally when planted at an altitude of 1000-2000 meters above sea level. For flowering and fruiting arabica coffee requires a dry period of 4-5 months a year [19]. Arabica coffee produces brews with very high cup quality, thus driving global market popularity and demand [20].

C. Coffee Bean Quality Standard

In general, coffee beans undergo a sorting and grading process based on established quality standards before being marketed. Quality testing on coffee beans is carried out by checking the physical quality and taste. Physical quality is determined based on the number of defects found in green beans based on Indonesian National Standard (SNI) Number 01-2907-2008 or the Specialty Coffee Association (SCA).

Coffee beans consist of two quality requirements, namely the general quality shown in **Table 1**. The quality of coffee beans is assessed based on the number of defect values from a 300 gram sample. The main provisions of SNI 01-2907-2008 regarding the quality of coffee beans, one of which is that based on the defect value, coffee is classified into 6 quality levels. For robusta, quality 4 is divided into sub-quality levels 4a and 4b.

TABLE I. COFFEE GRADES

Grades	Defect Totals
Grade 1	Defects maximum 11
Grade 2	Defect totals 12 to 25
Grade 3	Defect totals 26 to 44
Grade 4a	Defect totals 45 to 60
Grade 4b	Defect totals 61 to 80
Grade 5	Defect totals 81 to 150
Grade 6	Defect totals 151 to 225

III. Methodology

Real-time object detection aims to accurately predict the category and position of objects in an image with low latency. The YOLO series has been at the forefront of this research due to its balance between performance and efficiency. However, reliance on NMS and architectural inefficiencies have hindered optimal performance. YOLOv10 addresses these issues by introducing consistent dual assignment for NMS-free training and an overall accuracy-efficiency-driven model design strategy [21].

YOLOv10 is a state-of-the-art computer vision architecture designed for real-time object detection, which builds on the advancements of its predecessors. The YOLOv10 model achieves higher Average Precision (mAP) compared to previous YOLO models such as YOLOv9, YOLOv8, and YOLOv7 when compared to the COCO MS dataset [22].

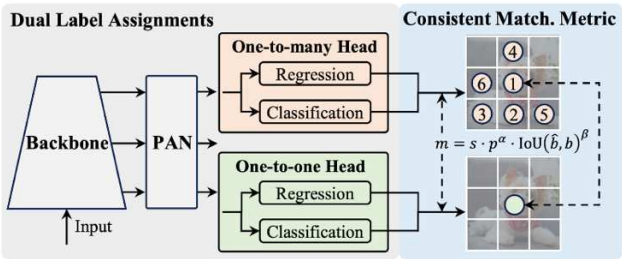

Figure 1. Consisten dual assignment for NMS-free training

Figure 1. Show the dual label assignments architecture of YOLOv10. The YOLOv10 architecture builds on the strengths of previous YOLO models by introducing several key innovations. The model architecture consists of the following components [21]:

1. **Backbone**, Responsible for feature extraction, the backbone in YOLOv10 uses an enhanced version of CSPNet (Cross Stage Partial Network) to improve gradient flow and reduce computational redundancy.
2. **Neck**, is designed to combine features from various scales and pass them to the head. It includes a PAN (Path Aggregation Network) layer for effective multiscale feature merging.
3. **One-to-Many Head**, Generate multiple predictions per object during training to provide rich supervision signals and improve learning accuracy.
4. **One-to-one Head**, Generates one best prediction per object during inference to eliminate the need for NMS, thereby reducing latency and improving efficiency.

IV. Experimental Result

A. Datasets Acquisition

The dataset was captured using a Sony Alpha A6000 camera equipped with a 35mm OSS lens, producing high-resolution images in JPG format with a 16:9 aspect ratio and a resolution of 6000x4000 pixels. A total of 700 images were taken, featuring samples of green Arabica coffee beans arranged specifically for defect detection. For each image, approximately 300 grams of coffee beans were randomly selected and placed on a white PVC mat, which served as a background to reduce noise during the feature extraction stage of modeling. Defective beans, categorized into two classes—brown beans and dried cherries—were strategically placed among the sample beans to simulate real inspection conditions and ensure accurate annotation for each defect class. The choice of a 300-gram sample aligns with industry standards for quality control, where this sample size is commonly used to represent a batch in quality assessments. The sample dataset is shown in **Figure 2**.

B. Data Prepocessings

The dataset utilized in this research comprises 700 images of green Arabica coffee beans, categorized into two primary classes: brown beans and dried cherries. The brown beans class includes beans with visual defects, such as brown coloring, which may indicate immaturity or inadequate drying processes. Conversely, the dried cherries class consists of beans that still retain dried fruit skin, indicating incomplete peeling, which is also considered a quality defect.

Figure 2. Sample dataset

Each image in the dataset manually annotated to specify the boundaries and labels of the objects in the images. This meticulous annotation process enables the model to detect and classify defects accurately. The image annotation data is stored in a *.txt file that includes information about the label, position coordinates (center-x & center-y), and size (width & height) of the bounding box in an image. Subsequently, the images were resized to a resolution of 1280x853 pixels to reduce file size and accelerate the training process. This process was successfully applied to all images in the dataset, resulting in 4.5 times smaller resolution and significantly reduced file size.

C. Result

Model development is used to conduct training by extracting features so that the device is able to recognize each image feature that has been labeled before. This model training, using the help of Google Collab.

Table II. Comparison Performance

	YOLOv10n	YOLOv8n
Epochs	100	100
Imgsize	640	640
Batch	32	32
Precision	0.985	0.993
Recall	0.986	0.994
mAP50	0.994	0.995
mAP50-90	0.866	0.873
Training time	0.479 hours	0.499 hours

Table II. Presents the performance comparison between several YOLO (You Only Look Once) models, namely YOLOv10n, YOLOv8n, in performing object detection. Each model was tested using 100 epochs, with an image size (imgsize) of 640 and a batch size of 32 to ensure equality in the test conditions.

The performance of the YOLOv8 and YOLOv10 models was evaluated based on standard object detection metrics, specifically Precision, Recall, mAP50, and mAP50-95. The Mean Average Precision at 50% Intersection over Union (mAP50) indicates the model's ability to detect objects with at least 50% overlap between predicted and actual bounding boxes. A higher mAP50 score signifies more accurate localization, which is crucial

in identifying minor defects that may otherwise go undetected.

Meanwhile, mAP50-95 is a more rigorous metric that averages the mAP across multiple IoU thresholds from 50% to 95%. This metric assesses the robustness of a model by evaluating its capacity to accurately detect objects even under conditions of low overlap. In this study, YOLOv8n achieved a mAP50 of 0.995 and a mAP50-95 of 0.875, while YOLOv10n achieved a mAP50 of 0.992 and a mAP50-95 of 0.859. The higher mAP50-95 score of YOLOv8n suggests superior detection under varying levels of object occlusion and overlap, a characteristic essential for high-stakes applications such as coffee defect detection, where precise differentiation between defective and high-quality beans is imperative.

In practical applications, particularly in real-time coffee bean sorting, a high mAP50 score ensures that the model reliably identifies defect locations on beans, even when they are partially visible or closely packed. Conversely, a high mAP50-95 score is advantageous in situations where fine-grained object detection is required, as it minimizes the likelihood of false positives and improves the model's confidence in distinguishing between slightly defective and non-defective beans. Given that YOLOv8n achieved marginally better results in both mAP metrics, it is potentially more suitable for deployment in industrial settings where accuracy in defect detection can significantly impact overall quality control processes.

Furthermore, YOLOv10n's advantage in training time (0.479 hours compared to YOLOv8n's 0.499 hours) suggests that YOLOv10n could be preferable in situations where computational resources are limited or when rapid model updates are necessary. However, the slight trade-off in accuracy should be considered based on the specific requirements of the coffee processing workflow.

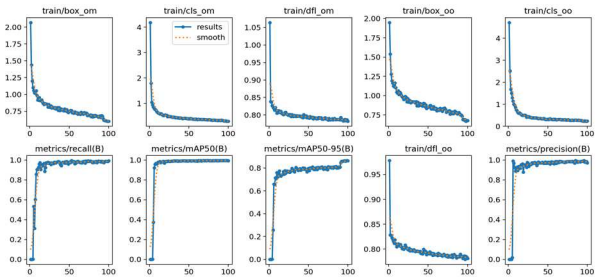

Figure 3. Graphis loss

Figure 3. Showing some evaluation metrics of the YOLO model training tracked for 100 epochs. The graphs at the top depict loss metrics, while the bottom shows the model's performance in terms of recall, precision, and mean average precision (mAP). Each graph has an x-axis representing the number of epochs, with a range from 0 to 100.

In the first row, there are loss metrics for different components of the model. The train/box_om and train/box_oo graphs show the loss in the bounding box, which decreases as the epochs increase, indicating an improvement in the accuracy of object location prediction. Similarly, train/cls_om and train/cls_oo represent the loss on object classification, which also decreases during training. This decrease indicates that the model is getting better at recognizing object classes. train/dfl_om and train/dfl_oo represent the DFL loss (distillation loss), which is related to the quality of bounding box prediction.

In the second row, the metrics/recall(B) graph shows a very rapid increase in recall at the beginning of training, and eventually stabilizes near 1 (100%), indicating that the model is increasingly able to detect most of the objects. metrics/mAP50(B) and metrics/mAP50-95(B) depict the model's ability to detect objects at various levels of IoU (Intersection over Union). Both metrics also improve significantly with training, although mAP50-95 is lower than mAP50.

The metrics/precision(B) graph shows that the model's precision also increases sharply at the beginning of training and eventually stabilizes near 1. This indicates that the model is very accurate in its predictions, with few detection errors. Overall, the trends seen from these graphs indicate that the model performed well in training, characterized by decreased losses and improved performance in object detection.

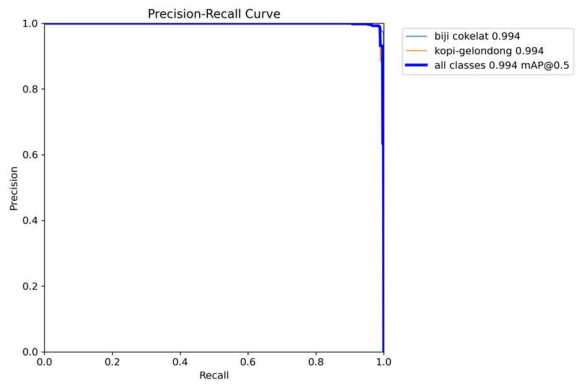

Figure 4. Precision curve

Figure 4. Shows the Precision-Confidence curve that illustrates the performance of the model in detecting objects based on the model's confidence in its predictions. On the X-axis, confidence indicates how confident the model is in the predictions made, while the Y-axis shows precision, which is the proportion of correct predictions out of all positive predictions. The light blue line represents the cocoa bean class, and the orange line represents the skunk coffee class, both of which show that with higher confidence, the precision is also closer to perfect.

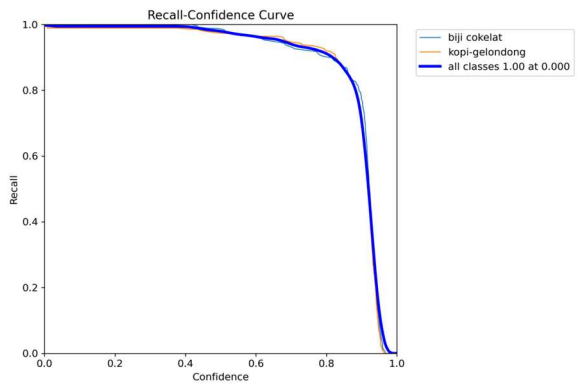

Figure 5. Recall curve

Figure 5. This Recall-Confidence Curve depicts the relationship between recall, which indicates how many

objects were detected by the model, and confidence, which indicates the confidence level of the model in its predictions. At low confidence, the recall is at a very high value, indicating that the model is able to detect almost all object instances of the chocolate bean and coffee-mondong classes. However, as the confidence increases, the recall starts to decrease significantly. The thick blue line representing the performance of all classes shows that the model achieves perfect recall at low confidence, but recall drops sharply when confidence reaches around 0.8 and above. This shows that although the model becomes more confident in its predictions, the number of successfully detected objects decreases as the model becomes more selective. As a result, the higher the confidence, the recall tends to drop drastically.

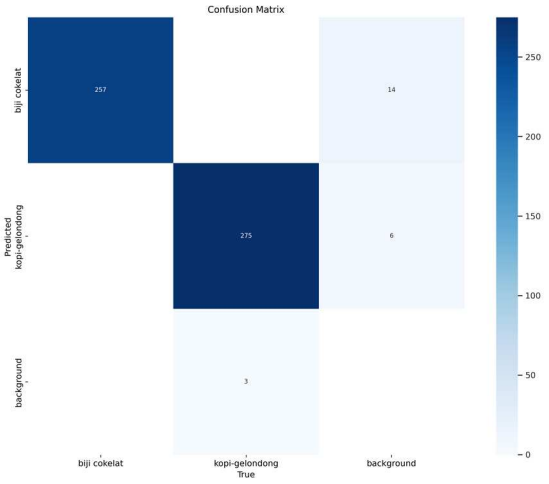

Figure 6. Confusion matrix

Figure 6. The confusion matrix shows the model's prediction results for three classes: cocoa, coffee-mondong, and background. For the cocoa bean class, the model correctly classified 257 samples, but erred on 14 samples, which were predicted as kopi-gelondong. Meanwhile, for the kopi-gelondong class, the model correctly classified 275 samples, but incorrectly predicted 6 samples as chocolate beans. In the background class, the model successfully identified 3 samples correctly with no mispredictions to other classes.

Overall, the model performed well in detecting cacao nibs and coffee-nibs, with most predictions being accurate. However, there were some errors between the two main classes (cocoa-nut and coffee-nut), indicating that the model had difficulty in distinguishing some samples from the two classes. The background class performed the best with only a few errors.

V. Conclusions

This study demonstrates the potential of YOLOv8 and YOLOv10 deep learning models for automating defect detection in green Arabica coffee beans, which is a critical component in maintaining quality standards in the coffee industry. Both models successfully identified defects across two classes, brown beans and dried cherries, achieving high accuracy and mAP scores. The results indicate that YOLOv8n, with its superior mAP50 and mAP50-95 scores, provides more accurate detection, making it ideal for applications where precise quality control is paramount. However, YOLOv10n offers a slight advantage in training time, which could be beneficial in settings with limited computational resources or where rapid model updates are required.

The findings suggest that integrating these models into real-time coffee sorting systems could significantly enhance the efficiency and consistency of quality control processes, reducing labor costs and minimizing human error. Future work could explore further optimization of these models for defect detection in various coffee bean types and consider additional environmental factors such as lighting and bean orientation. Additionally, experimenting with larger datasets and exploring more advanced versions of YOLO may yield further improvements in detection accuracy and processing speed.

Ultimately, the application of YOLO-based models for coffee bean defect detection represents a promising approach to addressing industry needs for scalable, reliable, and cost-effective quality assessment solutions.

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